

HAT-P-32b

R-1172

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_210111 · created 2026-07-31T03:27:09+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459225.7096 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1648 +/- 0.0053
Transit depth (Rp/Rs)^2	2.72 +/- 0.18 [%]
Orbital Inclination (inc)	84.01 +/- 0.5
Airmass coefficient 1 (a1)	1.0029 +/- 0.008
Airmass coefficient 2 (a2)	-0.0015 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	#3 - [622, 435]
Optimal Aperture	5.39
Optimal Annulus	11.16
Transit Duration (day)	0.1208 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1648 $\pm$ 0.0053 vs 0.1489 (deviation 3.0 σ)
Duration fit vs published	0.1208 d vs 0.1304 d (deviation 4.58 σ)
Mid-time O-C	-8.1 min
Depth SNR	31.1
Residual scatter	0.8 %

Light curve

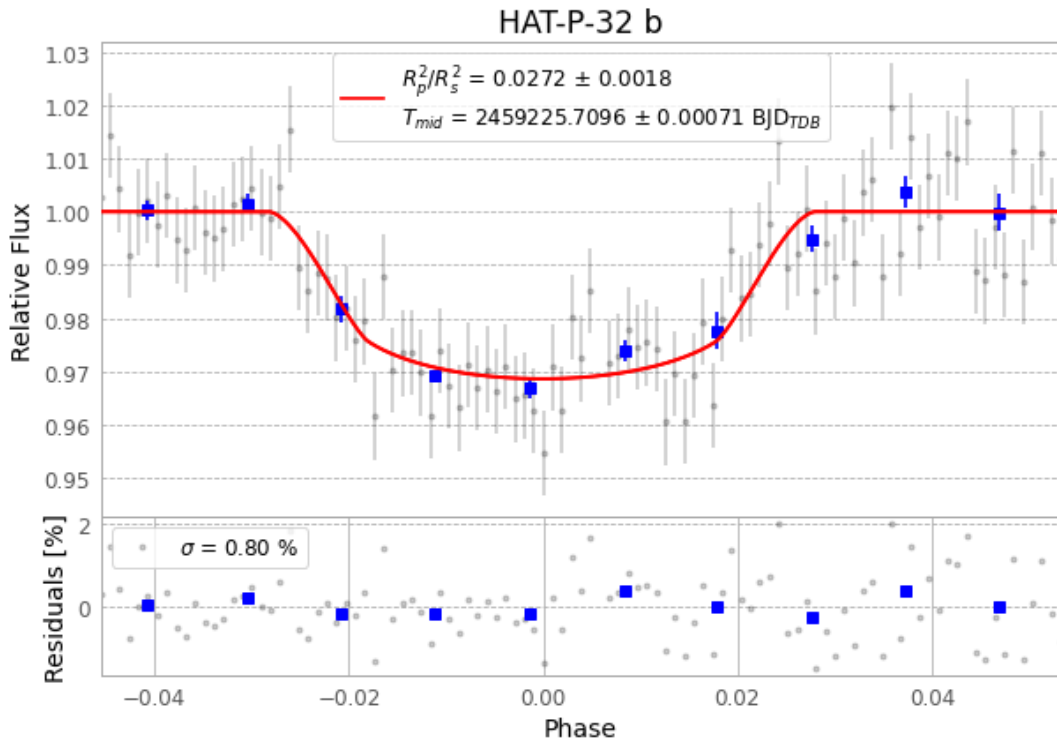


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-01-10.png (SHA-256 a4552f72fef96237...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [497, 356], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 769a3abcc6e066f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), HATP-32210111023612.FITS → HATP-32210111074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-210111.log (a7e573a81989...), FinalLightCurve_HAT-P-32 b_2021-01-10.png (a4552f72fef9...), FinalLightCurve_HAT-P-32 b_2021-01-10.csv (2841614d99e6...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:27Z.